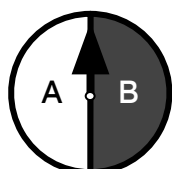


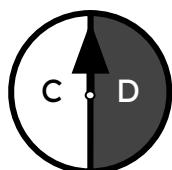


Using an outcome tree to display outcomes for more than two events

Task A: Outcome trees for multi-stage trials



spinner 1



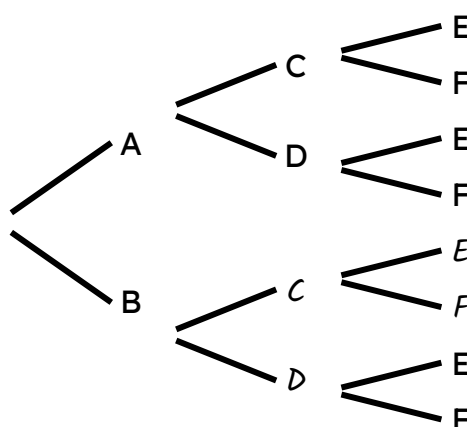
spinner 2



spinner 3

1) Sofia will spin spinner 1, 2, and 3 once each.

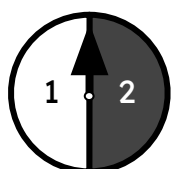
a) Complete this **outcome tree** and sample space.



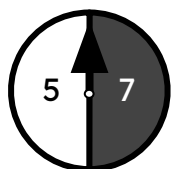
A C E
A C F
A D E
A D F
B C E
B C F
B D E
B D F

b) The event that at least one spinner lands on a vowel has how many outcomes?

There are six outcomes with at least one vowel.



spinner 1



spinner 2

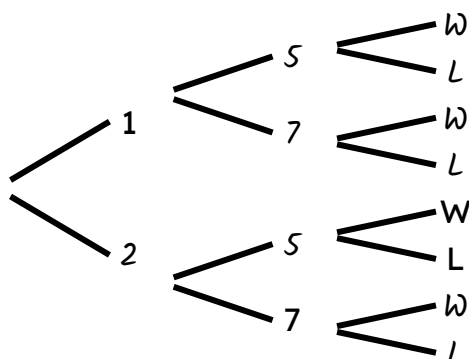


spinner 3

2) Aisha plays a game.
Aisha will spin spinners 1, 2, and 3 once each.

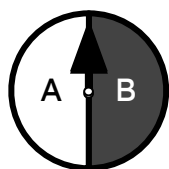
If spinner 3 lands on W, then Aisha wins the number of sweets equal to the **sum** of the numbers on spinners 1 and 2.

Complete this **outcome tree** and sample space for Aisha's game.

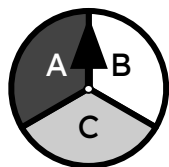


1 5 W
1 5 L
1 7 W
1 7 L
2 5 W
2 5 L
2 7 W
2 7 L

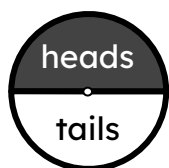
wins 6
no win
wins 8
no win
wins 7
no win
wins 9
no win



spinner 1



spinner 2

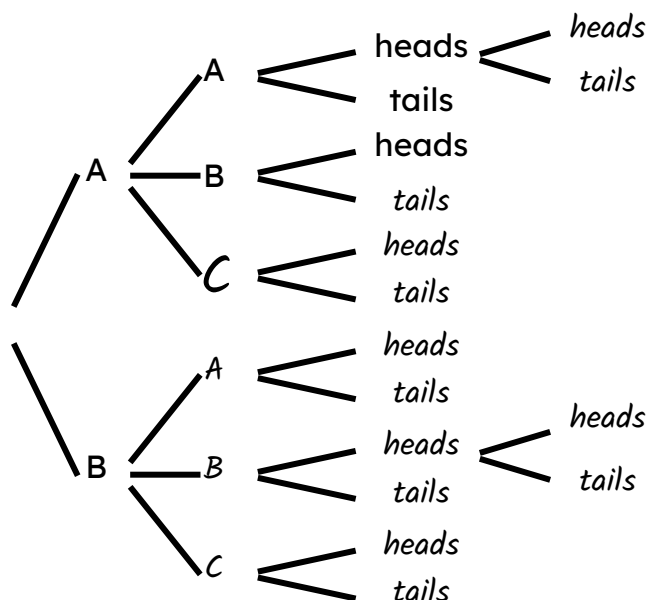


coin

3) Sam will spin spinners 1 and 2 once each, and then flip a coin.

a) Draw the **outcome tree** for this three-stage trial. Stage 1 is spinner 1, stage 2 is spinner 2, and stage 3 is the coin.

b) Modify this **outcome tree** to satisfy this four-stage trial: “if two spinners land on the same letter and the coin lands heads, then flip another coin”.

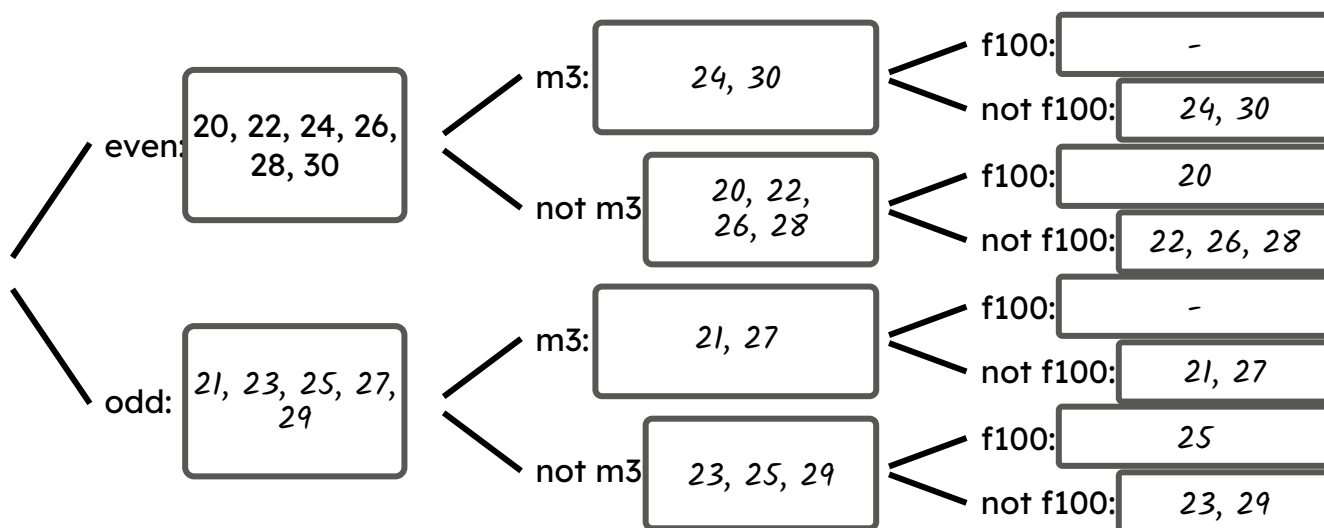


Task B: Outcome trees from multiple events

1) Sam will spin a spinner with the integers from 20 – 30 on it.

Sam considers the events that the spinner lands on an even number, a multiple of 3 (m3) and a factor of 100 (f100).

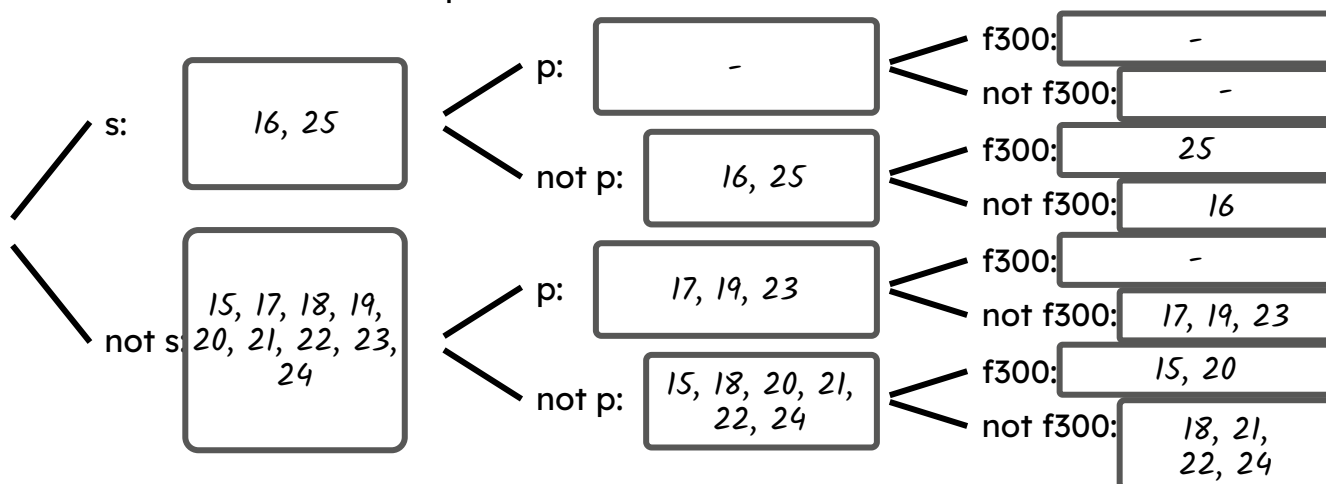
Fill in the blanks for this **outcome tree** with outcomes from this spinner.



2) Lucas will spin a spinner with the integers from 15 – 25 on it.

Lucas considers the events that the spinner lands on a square number (s), a prime number (p), and a factor of 300 (f300).

Fill in the blanks for and complete this **outcome tree**.



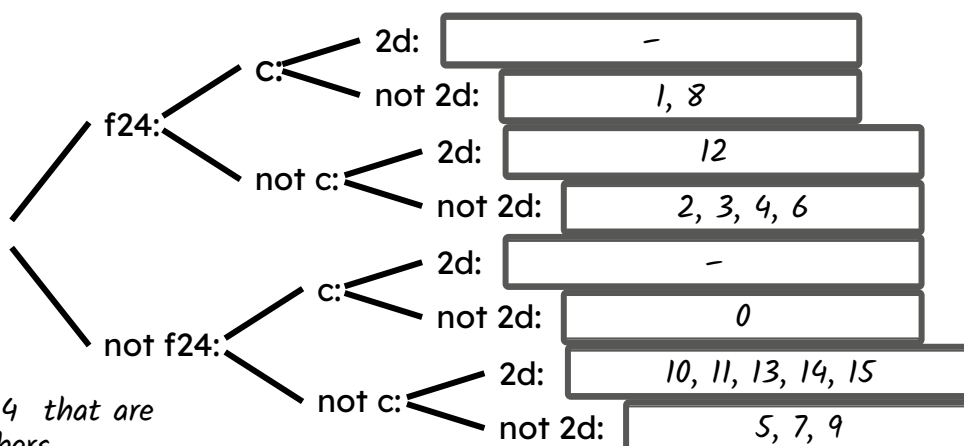
3) Jun will spin a spinner with the integers from 0 to 15 on it.

Jun considers the events that the spinner lands on a factor of 24 (f24), and a cube number (c), and a two-digit number (2d).

a) Complete the partial **outcome tree** and list the outcomes at the end of each branch.

b) How many outcomes match the events that Jun is looking out for?

There are zero factors of 24 that are two-digit cube numbers.



4) Sofia has a spinner with the following outcomes:
{1, 3, 4, 5, 8, 10, 12, 20, 25, 27, 36}

a) Construct and fill in a four-layer **outcome tree** for the outcomes of Sofia's spinner.

b) How many outcomes that are even numbers are also both square and triangular?

Only one: 36

